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NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE NAME

BIOLOGY CLASS

REGISTRATION NUMBER

# BIOLOGY

**9648/01**

Paper 1 Multiple Choice

**18th September 2014**

**1 hour 15 minutes**

Additional Materials: Multiple Choice Answer Sheet

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your Biology class, registration number and name above and on the Answer Sheet provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

1 Three pure substances are analysed. The table shows the elements that each contains.

| | C | H | O | N | P | S |
|---|---|---|---|---|---|---|
| **X** | + | + | + | + | - | - |
| **Y** | + | + | + | - | - | - |
| **Z** | + | + | + | + | - | + |

Which combination could be correct?

| | X | Y | Z |
|---|---|---|---|
| **A** | adenine | carbohydrate | fat |
| **B** | adenine | fat | an amino acid |
| **C** | an amino acid | fat | carbohydrate |
| **D** | fat | carbohydrate | an amino acid |

2 Each polypeptide chain of haemoglobin contains a number of regions in the form of an alpha helix.

Which feature of the haemoglobin molecule is responsible for this?

**A** Disulphide bonds between the four polypeptide chains at the quaternary structure.

**B** Hydrogen bonds between R groups of the amino acids in the secondary structure.

**C** Hydrogen bonds within the polypeptide chain in the secondary structure.

**D** Hydrophobic interactions within the polypeptide chain in the tertiary structure.

3 Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it as applied to the roots of the plant. As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

Which of the following statements are consistent with the scientist's observations?

**I** The toxin was stored in the central vacuole.
**II** The toxin cannot cross the membrane of the organelle in which it is stored.
**III** The toxin was stored in chloroplast.
**IV** The toxin is likely to be lipid-soluble.
**V** The toxin may be an enzyme.

**A** **I, II** and **V**
**B** **I**, **IV** and **V**
**C** **II**, **III** and **IV**
**D** **III**, **IV** and **V**

**4** The following statements describe three orders of structure of the HIV protease molecule.

**1** The dimeric enzyme consists of two identical subunits joined to one another.

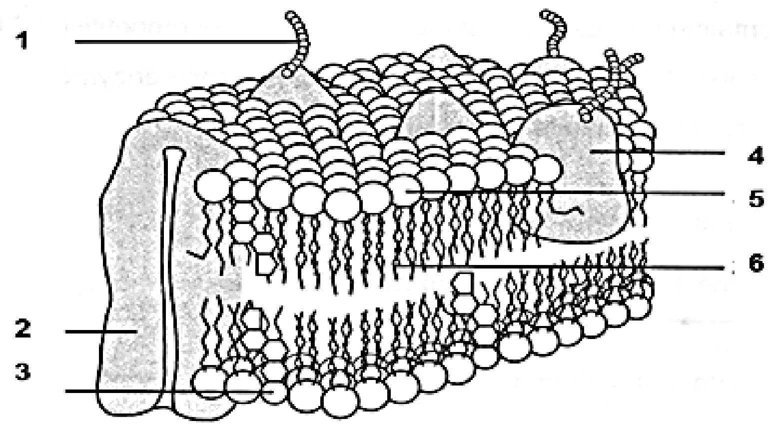
**2** The sequence of the 99 amino acids in each polypeptide chain is known.

**3** The amino acids in each chain, either coiled into helices or folded into beta sheets, are held in position by hydrogen bonds.

Which order is described by each statement?

| | **Statement 1** | **Statement 2** | **Statement 3** |
|---|---|---|---|
| **A** | Quaternary | Primary | Tertiary |
| **B** | Secondary | Primary | Quaternary |
| **C** | Tertiary | Primary | Secondary |
| **D** | Quaternary | Primary | Secondary |

**5** The diagram shows part of a cell surface membrane.




What is the correct function for each of the structures labelled?

| | Regulates membrane fluidity | Involved in cell-cell recognition | Transports ions and large polar molecules |
|---|---|---|---|
| **A** | 3 | 1 | 2 |
| **B** | 6 | 1 | 2 |
| **C** | 3 | 5 | 1 |
| **D** | 6 | 3 | 2 |

**6** Some inhibitors of enzyme reactions bind to the enzyme-substrate complex. Which statements about this type of inhibition are correct?

1. The active site changes shape.
2. The inhibitor is non-competitive.
3. The initial rate of reaction is reduced.
4. The maximum rate of reaction (Vmax) is increased.

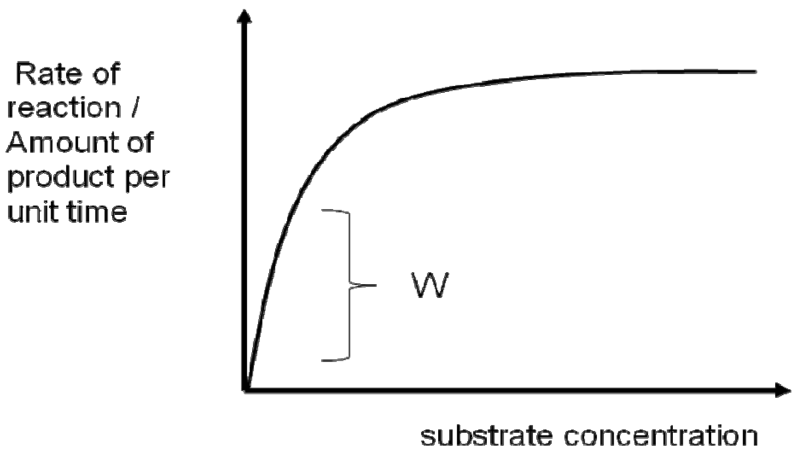
**A** 1 and 2 only

**B** 1 and 3 only

**C** 2 and 3 only

**D** 2, 3, and 4 only

**7** The graph below shows the rate of an enzyme catalyzed reaction occurring in lysosome with increasing substrate concentration. The reaction is carried out at 37°C and a pH of 4 for all substrate concentrations.




Which of the following(s) would result in a ***decrease*** in the rate of reaction at W?

1 Addition of co-factor

2 Decrease in temperature to 27°C

3 Increase in pH to 9

4 Addition of competitive inhibitor

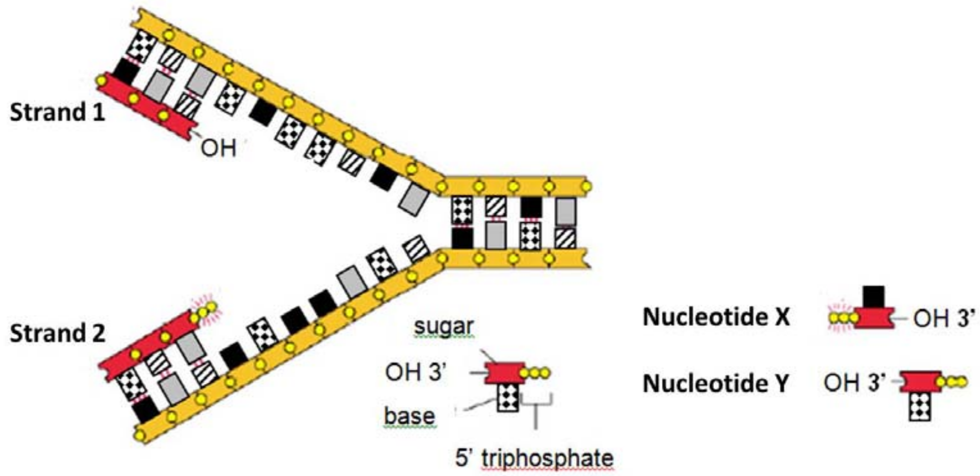
**A** 1 and 4 only

**B** 2 and 3 only

**C** 2 , 3 and 4 only

**D** 1, 2, 3 and 4

8 DNA replication is illustrated in the figure.




Which of the following correctly describes the addition of the next nucleotide(s) to the DNA strands undergoing replication?

**A** Nucleotide **X** will be added to the leading strand, which is strand 1.

**B** Nucleotide **Y** will be added to the leading strand, which is strand 1.

**C** Nucleotide **X** will be added to the lagging strand, which is strand 1.

**D** Nucleotide **Y** will be added to the leading strand, which is strand 2.

9 The diagram shows part of the normal sequence of an mRNA molecule.

CCAAGUGGUCCGCUAAGAAGGC

A mutation in the DNA resulted in a polypeptide beginning with the following sequence.

glycine - serine - proline - glycine - isoleucine - leucine

The DNA triplets for some amino acids are

| Glycine | Isoleucine | Leucine | Proline | Serine |
|---|---|---|---|---|
| CGA | ATA | TTA | CCA | TCA |
| GGT | ATT | CTT | CCG | TCG |
| GGC | | CTC | | |

Which mutation has occurred in the DNA molecule?

**A** A reversal in the order of nucleotide

**B** An addition of an extra nucleotide

**C** The loss of a nucleotide

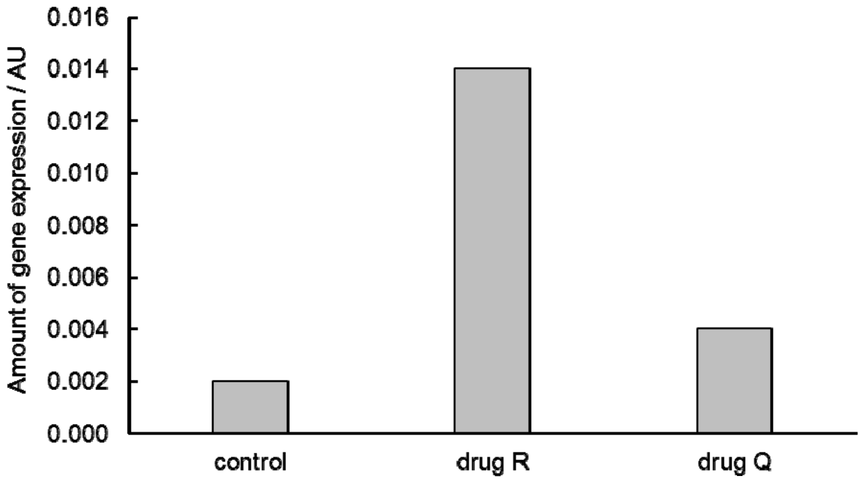
**D** The replacement of one nucleotide by a different nucleotide

**10** A double-stranded DNA molecule is 10 kb long.

Which row correctly describes the structure of this DNA molecule?

| | **number of nucleotides** | **length of DNA molecule / μm** | **number of complete turns** |
|---|---|---|---|
| **A** | 10000 | 3.4 | 1000 |
| **B** | 20000 | 0.34 | 1000 |
| **C** | 20000 | 3.4 | 1000 |
| **D** | 20000 | 3.4 | 10000 |

**11** Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drug R and Q on expression of a gene. The graph shows the experimental results.




Which are possible explanations to the results shown?

1 Drug Q results in weaker binding of histones to DNA.

2 Drug Q increases gene expression by increasing accessibility of RNA polymerase to the promoter.

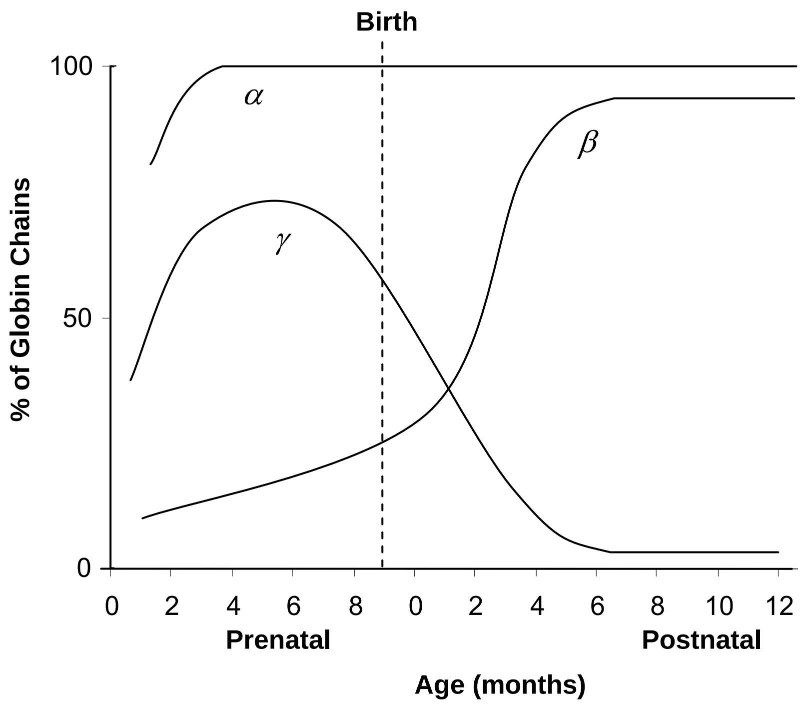
3 Drug R increases gene expression by preventing methylation at CpG islands at the promoter.

4 Inhibiting DNA methylation is more effective in increasing gene expression than inhibiting histone deacetylation.

**A** 1 and 2 only **B** 2 and 4 only **C** 1, 2 and 3 only **D** All of the above

**12** The globin gene family in humans consists of the α, β and γ genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.




Which of the following cannot account for the differences in the levels of expression of globin chains?

**A** Methyl groups are added to regulatory sequences of *γ*-globin genes during the postnatal period, allowing for some proteins to bind.

**B** Alternative splicing occurs in the mature mRNA of the *α*-globin and *β*-globin genes, resulting in differences in the rate of expression of globin chains during the prenatal period.

**C** A growth factor triggers the expression of a transcription factor that increases the rate of *β*-globin gene expression during the postnatal period.

**D** The shortening of poly(A) tail in the mRNA of *γ* -globin genes reduces its stability, resulting in a decrease in the rate of expression of *γ*-globin chains during the postnatal period.

**13** S and T are two identical cells. Only the X chromosomes are shown in these cells. When these cells undergo meiosis to form gametes, non-disjunction of the chromosomes occurred. In cell S, it occurred during meiosis I whereas in cell T, it occurred during meiosis II.

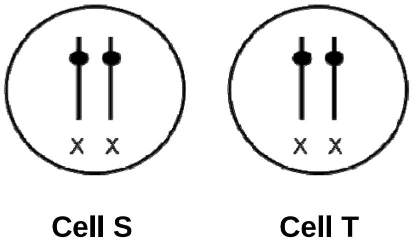
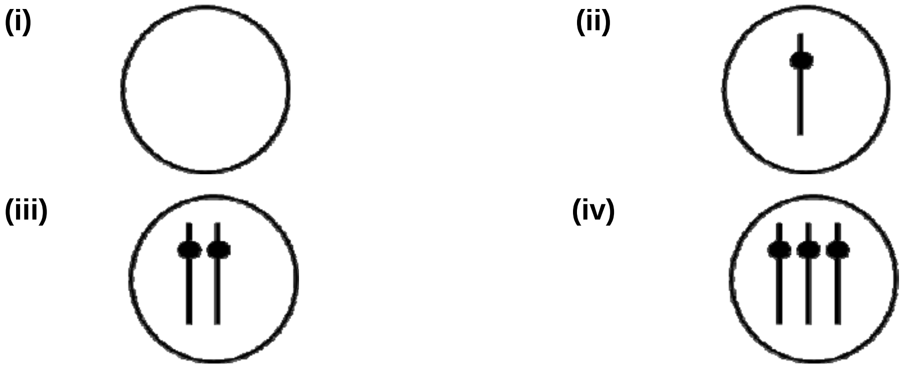



*Diagram above showing 2 Identical cells, S and T*

The diagrams below show some combination of X chromosome(s) in gametes.




Which of the following options shows correctly the type of gametes that is/are common to both form of non-disjunction?

**A** (ii) only.

**B** (ii) and (iii) only.

**C** (i) and (iii) only.

**D** (ii) and (iv) only.

**14** How many different types of gametes would be produced by an organism of genotype PpqqRrSsTTUu, if all of the genes assort independently?

**A** 8

**B** 16

**C** 32

**D** 64

**15** Most cancers are thought to originate from a single cell which has undergone a change in its genetic aberrations. In these cases a defect in the genome is thought to predispose the cell to malignant transformation, which then takes place only after further genetic disturbance. Tumour progression, whereby a benign tumour becomes a malignant one, is accelerated by mutagenic agents (tumour initiators) and nonmutagenic agents (tumour promoters). Both types of agent affect gene expression, stimulate cell proliferation, and alter balance of mutant and non-mutant cells.

The following tables show the relative risks (RR) of tobacco and alcohol use for cancer of the endolarynx, epilarynx, and hypopharnyx. A relative risk greater than 1.0 indicates increased risk relative to a control group.

Based on the passage, what is the direct role of tumour initiators?

**A** They will cause changes in the DNA sequence of a cell

**B** They might cause noncancerous cells to become cancerous

**C** They could change the metastatic potential of a cell

**D** They could cause tumours to develop in normal tissue

**16** Below are some statements related to cancer:

Which of the following statements are true?

1. Oncogenes and tumor suppressor genes can both be detected by introducing fragmented DNA from cancer cells into suitable cell lines and isolating colonies that display cancerous properties
2. Individuals who inherit one inactive copy of tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies
3. Viruses and other infectious agents play no role in human cancers
4. In the cellular regulatory pathways that control cell growth and proliferation, the products of oncogenes are inhibitory components and the products of tumour suppressor genes are stimulatory components
5. When analysed, cancer cells are often found to have only one mutation in a regulatory pathway that controls cell proliferation

**A** 1 and 2

**B** 1, 2, and 3

**C** 1, 3, and 5

**D** 1, 2, and 4

**17.** The proteins and glycoproteins which make up the capsid, envelope and spikes of a virus determine the infective properties of that virus. Which of the following is not true?

**A** Some complex viruses replicate without the synthetic machinery of the host cell

**B** Animal viruses enter their host via endocytosis

**C** A bacteriophage shed its protein coat outside the host cell and injects its nucleic acids through the host cell wall

**D** A latent bacteriophage consisting only of a DNA fragment is called a prophage.

**18.** Which of the following is true concerning animal retroviruses?

**A** They must replicate during the S phase of the cell cycle.

**B** They require an RNA-dependent DNA polymerase

**C** The virions have double-stranded RNA genomes

**D** Replication of their genome occurs entirely within the host nucleus

**19.** In *Escherichia coli*, the synthesis of tryptophan is controlled by the tryptophan operon that is repressed in the presence of excessive tryptophan. When a mutant strain that has lost the regulatory gene of tryptophan operon is placed in a medium that contains all nutrients the cell need to grow except tryptophan, which of the following will occur?

**A** the cells will continue to grow even though excess tryptophan is synthesized

**B** The cells will grow until excessive tryptophan arrests the expression of the operon

**C** The cells will not grow until enough tryptophan has been synthesised to active the corepressor

**D** The cells will never grow unless tryptophan is added to the medium

**20.** Catabolite repression in *E. coli* bacteria, involving the catabolite activator protein (CAP), is actually a type of positive regulation. Which of the following is a possible explanation?

**A** cAMP-CAP helps RNA polymerase to initiate transcription

**B** allolactose -CAP prevents RNA polymerase from initiating transcription

**C** glucose binds to CAP and activates it

**D** CAP complex binds to operator and increases transcritption

**21** Which of the following statements are **not** true?

i) The carbon dioxide taken in by plants is the source of the oxygen evolved in photosynthesis.
ii) Both the mitochondria and the chloroplast are responsible for the production of cellular ATP.
iii) Photosystem I can operate independently from photosystem II.
iv) At low light intensities, energy for triose phosphate production is supplied by ATP produced by respiration.
v) The proton motive force is only generated in the presence of light and oxidized co-enzymes.

**A** i, ii, iii only

**B** i, ii , iv and v only

**C** ii, iii and v only

**D** iii, iv and v only

**22** The table below shows reactions occurring in a plant cell, and their respective locations.

| | **Reaction** | **Location in a cell** |
|---|---|---|
| **1.** | ribulose bisphosphate + $CO_2$ → glycerate-3-phosphate | Stroma |
| **2.** | glucose + ATP → glucose-6-phosphate + ADP | Matrix |
| **3.** | oxygen + $4H^+$ + $4e^-$ → 2 $H_2O$ | Stroma |
| **4.** | oxaloacetate + acetyl-CoA → citrate | Matrix |

Which of the following is / are **incorrectly** matched?

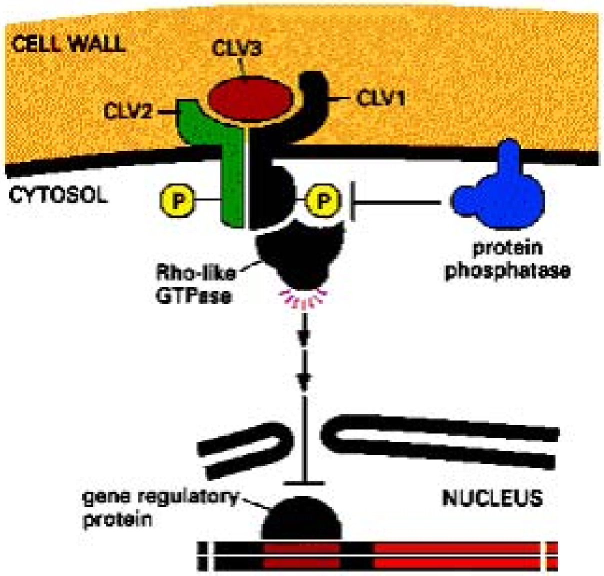
**A** **2** only

**B** **4** only

**C** **2** and **3** only

**D** **1**, **2** and **3**

**23** The diagram below shows the cell signalling pathway involved in cell proliferation and differentiation in the shoot meristem.




Legend:
—┤ Inhibition

Which one of the following sequences regarding the activation of the above signalling pathway is **correct**?

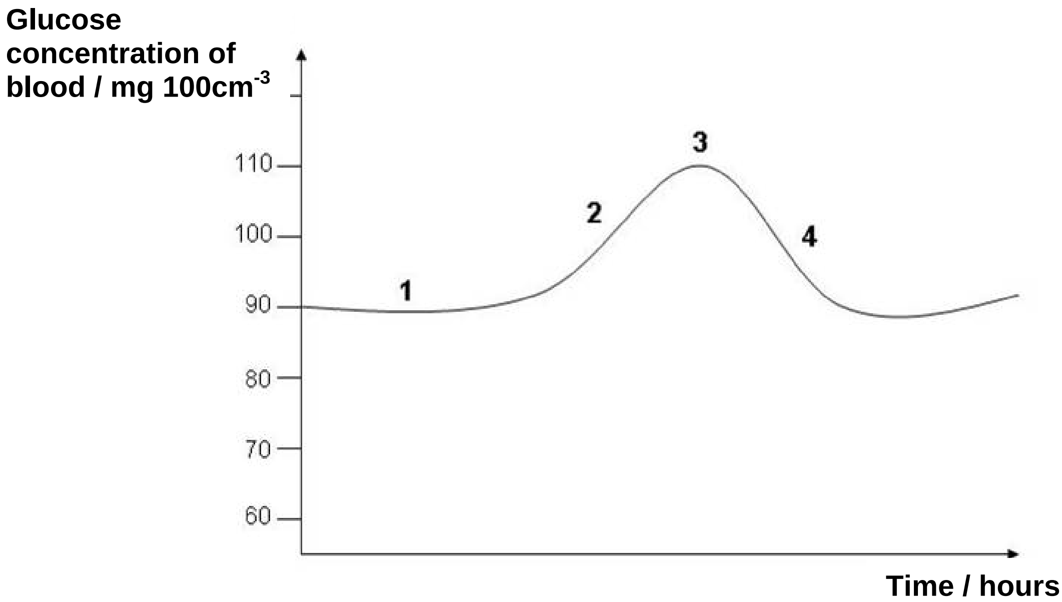
**A** binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → inhibition of gene transcription

**B** dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription

**C** binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription.

**D** dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription

**24** The graph below shows the changes of blood glucose level in a normal healthy person who consumes a meal.

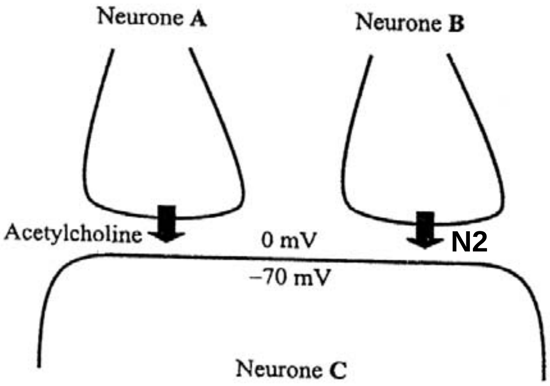



What is represented at each stage?

| | Negative feedback | Reference point | Effector cells activated |
|---|---|---|---|
| **A** | 4 | 1 | 3 |
| **B** | 4 | 1 | 2 |
| **C** | 3 | 2 | 4 |
| **D** | 3 | 4 | 2 |

**25** The diagram below shows axon terminals from each of two neurones, **A** and **B**, forming synapses with a third neurone, **C**.

When stimulated, neurone **A** releases acetylcholine while neurone **B** releases a structurally different neurotransmitter, N2. Concentrations of some ions are also shown.




| | Ion concentration / mmol dm$^{-3}$ | | |
|---|---|---|---|
| | **$Na^+$** | **$K^+$** | **$Cl^-$** |
| Outside neurone C | 145 | 12 | 120 |
| Inside neurone C | 12 | 155 | 5 |

Which of the following statements best explains why an action potential is not triggered in neurone **C** when both neurones **A** and **B** are stimulated?

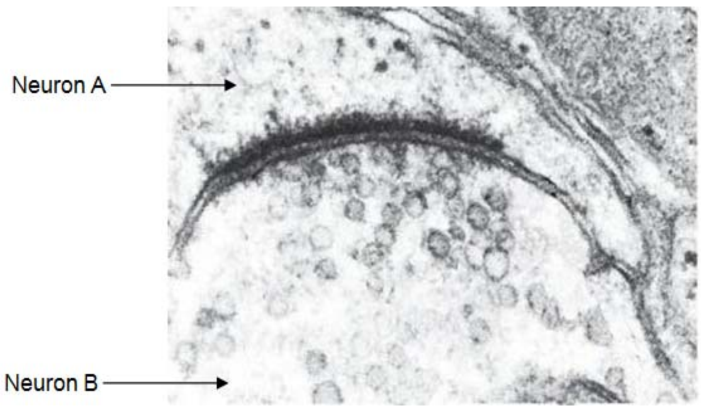
**A** N2 competes with acetylcholine for receptor binding at the postsynaptic membrane of neurone **C**, resulting in less $Na^+$ ions entering to cause EPSP.

**B** N2 causes the membrane potential of neurone **C** to be less negative, therefore more acetylcholine must be released by neurone **A** to cause greater influx of $Na^+$ ions.

**C** N2 increase the permeability of the postsynaptic membrane of neurone C to $Cl^-$ ions, resulting in IPSP.

**D** Acetylcholine is released slowly from neurone **A** while N2 is released rapidly from neurone **B**.

**26** The micrograph below shows a synapse between two neurons.




Which of the following statements about synaptic transmission is correct?

**A** Neuron B has received the synaptic vesicles from Neuron A.

**B** The membrane of Neuron A has many neurotransmitter channels.

**C** Calcium influx is triggered by depolarization that is produced by the arrival of an action potential in Neuron A.

**D** Neurotransmitters triggers voltage-gated sodium ion channels in Neuron A.

**27** When a pure-breeding maize plant with yellow seeds and full endosperm is crossed with another pure-breeding plant having white seeds and shrunken endosperm, the $F_1$ maize plants are found with yellow seeds and full endosperm. When the $F_1$ plants are selfed, four classes of descendants are obtained as shown in the table below:

| Colour of seed | Shape of endosperm | Number of seeds observed in $F_2$ generation (O) |
|---|---|---|
| Yellow | Full | 128 |
| Yellow | Shrunken | 27 |
| White | Full | 20 |
| White | Shrunken | 17 |

The formula for the chi-squared ($\chi^2$) test is given as follows:

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

where $\Sigma$ = 'sum of'
O = observed 'value'
E = expected 'value'

| Degrees of freedom | Probability | | | |
|---|---|---|---|---|
| | **0.10** | **0.05** | **0.01** | **0.001** |
| **1** | 2.71 | 3.84 | 6.64 | 10.83 |
| **2** | 4.69 | 5.99 | 9.21 | 13.82 |
| **3** | 6.25 | 7.82 | 11.35 | 16.27 |
| **4** | 7.78 | 9.49 | 13.28 | 18.47 |

Which of the following conclusions made about the inheritance of colour of seed and shape of endosperm is **correct**?

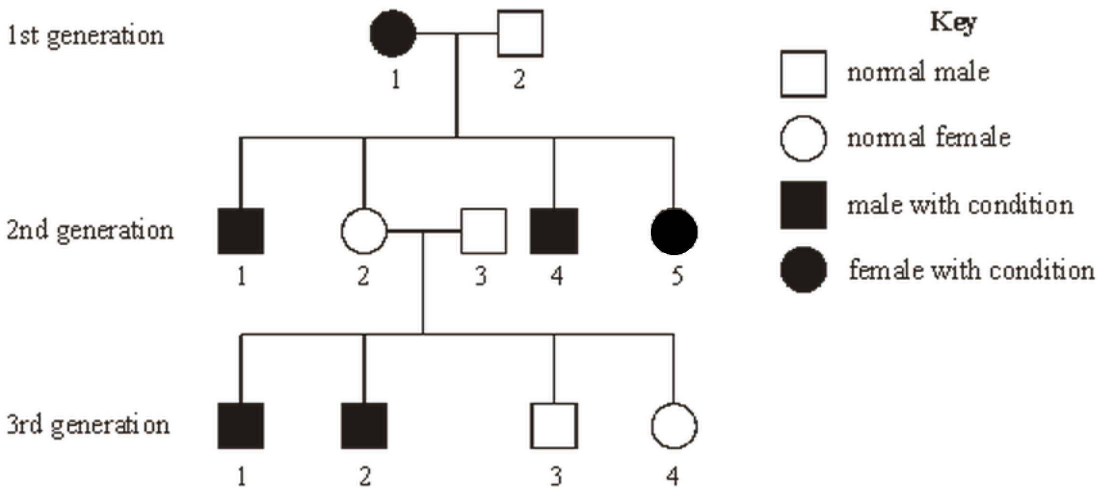
**A** Since $p < 0.05$, the difference between the observed and expected results is not significant. Thus the inheritance of colour of seed and shape of endosperm is following Mendel's law of independent assortment.

**B** Since $p > 0.05$, the difference between the observed and expected results is not significant. Thus the inheritance of colour of seed and shape of endosperm is not following Mendel's law of independent assortment.

**C** Since $p < 0.01$, the difference between the observed and expected results is not significant. Thus the inheritance of colour of seed and shape of endosperm is following Mendel's law of independent assortment.

**D** Since $p < 0.01$, the difference between the observed and expected results is significant. Thus the inheritance of colour of seed and shape of endosperm is not following Mendel's law of independent assortment.

**28** The diagram below is the pedigree of a family with a disease condition.




If individual 4 from the 3rd generation marries a normal male, what is the probability that the first offspring is a normal male?

**A** 1/4 **B** 1/3 **C** 1/2 **D** 3/8

**29** Two true breeding *Drosophila* are crossed: a normal- winged, red-eyed female and a miniature-winged, vermilion-eyed male. The F1 all have normal wings and red eyes. F1 offspring are crossed with miniature-winged, vermilion-eyed flies. The following offspring were obtained:

233 normal wing, red eye
247 miniature wing, vermilion eye
7 normal wing, vermilion eye
13 miniature wing, red eye

From these data, one could conclude that the alleles for miniature wings and vermilion eye are

**A** both X-linked and dominant.

**B** located on autosomes and dominant.

**C** recessive, and these genes are located 4 map units apart.

**D** recessive, and the deviation from the expected 9:3:3:1 ratio is due to epistasis.

**30** Tetrodotoxin, a puffer fish toxin, blocks voltage-gated sodium channels. Black widow spider's venom causes the voltage-gated calcium channels to be constantly open. Crotoxin binds irreversibly to acetylcholine receptors. What will happen to the nerve transmission if each toxin is applied?

| | **Tetrodotoxin** | **Black widow spider's venom** | **Crotoxin** |
|---|---|---|---|
| **A** | block action potentials along axon | reduce transmission of impulse across synapse | increase transmission of impulse across synapse |
| **B** | increase transmission of impulse across synapse | reduce transmission of impulse across synapse | block action potentials along axon |
| **C** | block action potentials along axon | increase transmission of impulse across synapse | reduce transmission of impulse across synapse |
| **D** | reduce transmission of impulse across synapse | block action potentials along axon | increase transmission of impulse across synapse |

**31** Female mites of the species *Pyemotes tritici* inject an extremely potent venom into their prey and able to paralyze insects 150 000 times their size. The DNA coding for three different polypeptide components of this toxin has been cloned and incorporated into a virus which only infect the insects. These genes are expressed when the recombinant virus infects an insect, thus reduce crop losses from pest.

Which of the following statement is true?

**A** There is no ethical concern as the virus only infects insects.

**B** The virus can be detrimental to the ecosystem if it harms non-targeted insect species.

**C** As the genes are present in the virus, there will be low chance of insects developing resistance due to possibility of mutation of the genes in virus.

**D** The toxin will never be detrimental to humans as the toxin will only be expressed upon viral infection on an insect.

**32** In a laboratory-based experiment, three groups of larvae of caddis fly, *Helicopsyche borealis*, were fed on pollen containing:

**Group A** no Bt toxin
**Group B** Bt toxin at concentrations found in streams in maize-growing area
**Group C** Bt toxin at concentrations twice as high as found in those streams

The researchers measured the mortality rates of the caddis fly larvae. Their results are summarised below.

| Groups compared | Difference in mortality rate |
|---|---|
| Groups A and B | No significant difference |
| Groups A and C | Significant greater mortality in C than in A |

The researchers independently made the following suggestions:

1. Bt toxin in the streams probably came from Bt maize growing around the region
2. Bt toxin concentration found in the stream is lethal to the caddis fly
3. Bt toxin concentration when in twice the amount is lethal to the caddis fly
4. Bt toxin is lethal to any organism that consumes water from the stream
5. There is no chance that Bt gene can be transferred to other organisms in the vicinity

Which of the above suggestions is/are true?

**A** 2 and 3 only

**B** 2, 3, and 5 only

**C** 1 and 3 only

**D** 1, 2, 3 and 4 only

**33** Which of the following statements about evolution is correct?

**A** Lamarck's theory of use and disuse adequately describes why giraffes have long necks.

**B** Darwin's theory of natural selection relies solely on genetic mutation

**C** Natural selection is the process in which random mutations are selected for survival by the environment

**D** Darwin's theory explains the evolution of man from present day apes

**34** Why do we think that molecular clocks exist and are accurate enough to be informative?

**A** This is because theory tells us that the rate of fixation of neutral alleles should be different in small and in large populations

**B** Data on protein evolution shows that protein molecules change at similar rates in different lineages

**C** We can measure the rate of fixation of neutral alleles in modern populations and extrapolate to the distant past

**D** Ages of lineages inferred from molecular clocks are often quite different from ages inferred from fossils

**35** One therapy to treat β-thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β-globin to the patient's cells.

Which of the following would ensure that the normal gene is passed on to the next generation?

**A** Using a retrovirus to introduce the normal β-globin gene into bone marrow cells.

**B** Using a retrovirus to introduce the normal β-globin gene into an egg cell.

**C** Using an adenoviral vector to introduce the normal β-globin gene into bone marrow cells.

**D** Using an adenoviral vector to introduce the normal β-globin gene into an egg cell.

**36** Which of the following statement about stem cells is correct?

**A** Stem cells are specialised cells that have yet to express the genes and proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.

**B** Stem cells are fully differentiated cells that reside under the surface of epithelia, in position to take over the function of the tissue when the overlying cells become damaged or wore out.

**C** Stem cells are undifferentiated embryonic cells that persist in the adult, and can give rise to all the cell types in the body.

**D** Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter that remains a stem cell, and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.

**37** Non-viral *ex vivo* transfer of a gene encoding coagulation factor VIII was performed using stem cells isolated from the bone marrow of patients suffering from severe hemophilia A.

What is the sequence of events for the *ex vivo* transfer of the gene encoding coagulation factor VIII?

**I** Transfection with vectors containing the gene encoding coagulation factor VIII

**II** Implantation of cells into patients

**III** Isolation of stem cells

**IV** Selection and cloning of cells expressing coagulation factor VIII

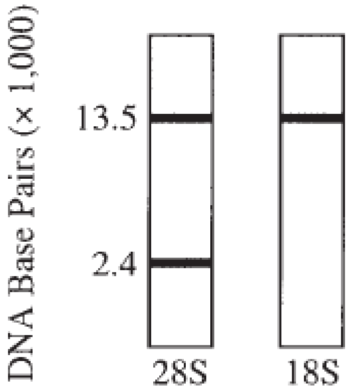
**A** **II, III, I, IV**

**B** **III, I, IV, II**

**C** **III, II, IV, I**

**D** **III, IV, II, I**

**38** The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA. In the autoradiogram on the left, the probe was 28S rRNA; at the right, the probe was 18S rRNA.

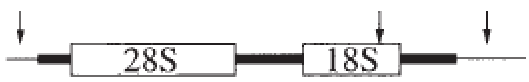



If the arrows in the following map show the location of the restriction sites of this restriction enzyme, which map *best* explains the results shown above?

**A**

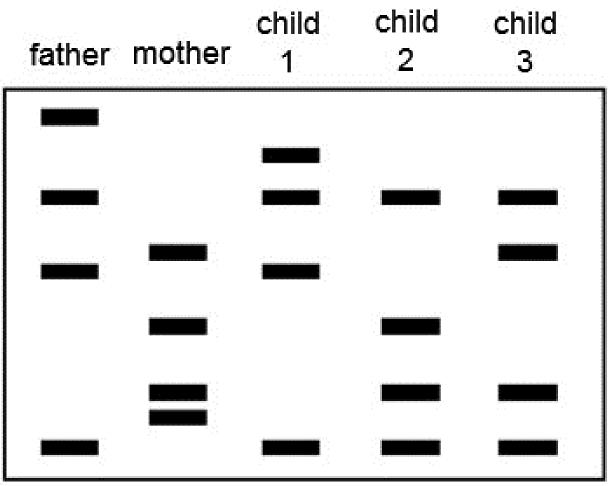

**B**


**C**




**D**

39 DNA fingerprinting is often used to confirm the identity of an individual. The diagram shows a small section of the gel electrophoresis results from a DNA fingerprinting analysis of a family.




Which of the children is least likely to be the offspring of both parents?

**A** child 1

**B** child 2

**C** child 3

**D** all of the children are definitely offspring of both parents

40 Which of the following DNA sequences would a restriction enzyme usually recognize and cut?

**A** 5' – ATGCAC – 3'

**B** 5' - GATATC – 3'

**C** 5' - TAGATA – 3'

**D** 5' - AATATA – 3'

**End of 2014 Prelim paper 1**